TEEGALA KRISHNA REDDY ENGINEERING COLLEGE

PROJECT TITLE:

SMART TIME TABLE GENERATOR (STTG)

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Abstract

The Smart Time Table Generator (STTG) addresses the labor-intensive and error-prone nature of manual institutional scheduling by providing an automated, deterministic solution. Traditional methods often result in double-booked faculty or rigid workloads. By utilizing a Constraint Satisfaction Problem (CSP) algorithm with backtracking logic, the system systematically explores scheduling parameters to guarantee a collision-free timetable. The platform strictly enforces hard constraints, such as room capacity and double-booking, while optimizing soft constraints like faculty preferences. Beyond initial generation, the STTG features a rules-based smart substitution module that automatically queries the database to identify the best available faculty based on subject expertise and current workload.

INTRODUCTION

Traditional timetabling is a cumbersome, manual process that costs institutions valuable time and resources, often leading to scheduling conflicts and faculty burnout. The Smart Time Table Generator (STTG) aims to deliver a collision-free, optimized schedule rapidly using a deterministic Constraint Satisfaction Problem (CSP) algorithm with backtracking logic. By automating these complexities through structured constraints rather than manual guesswork, the system shifts institutional scheduling from reactive management to proactive optimization, significantly reducing administrative overhead and improving overall academic planning efficiency.

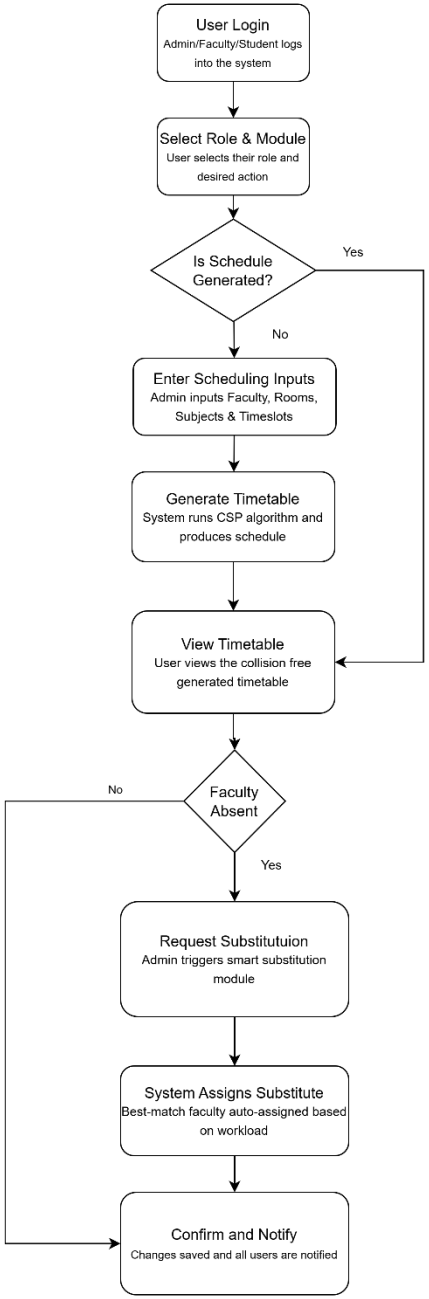
Existing System

Current institutional scheduling systems rely heavily on manual processes, where administrators spend weeks creating and finalizing timetables that are highly susceptible to human error, often resulting in double-booked rooms, faculty overlaps, and uneven workload distribution. Reshuffling schedules due to unexpected faculty absences is an incredibly difficult and time-consuming task, frequently leaving departments without adequate coverage. These rigid, error-prone methods lead to imbalanced lecture distributions among faculty and provide no systematic mechanism for conflict detection or automated substitution, ultimately causing significant inefficiencies in academic operations.

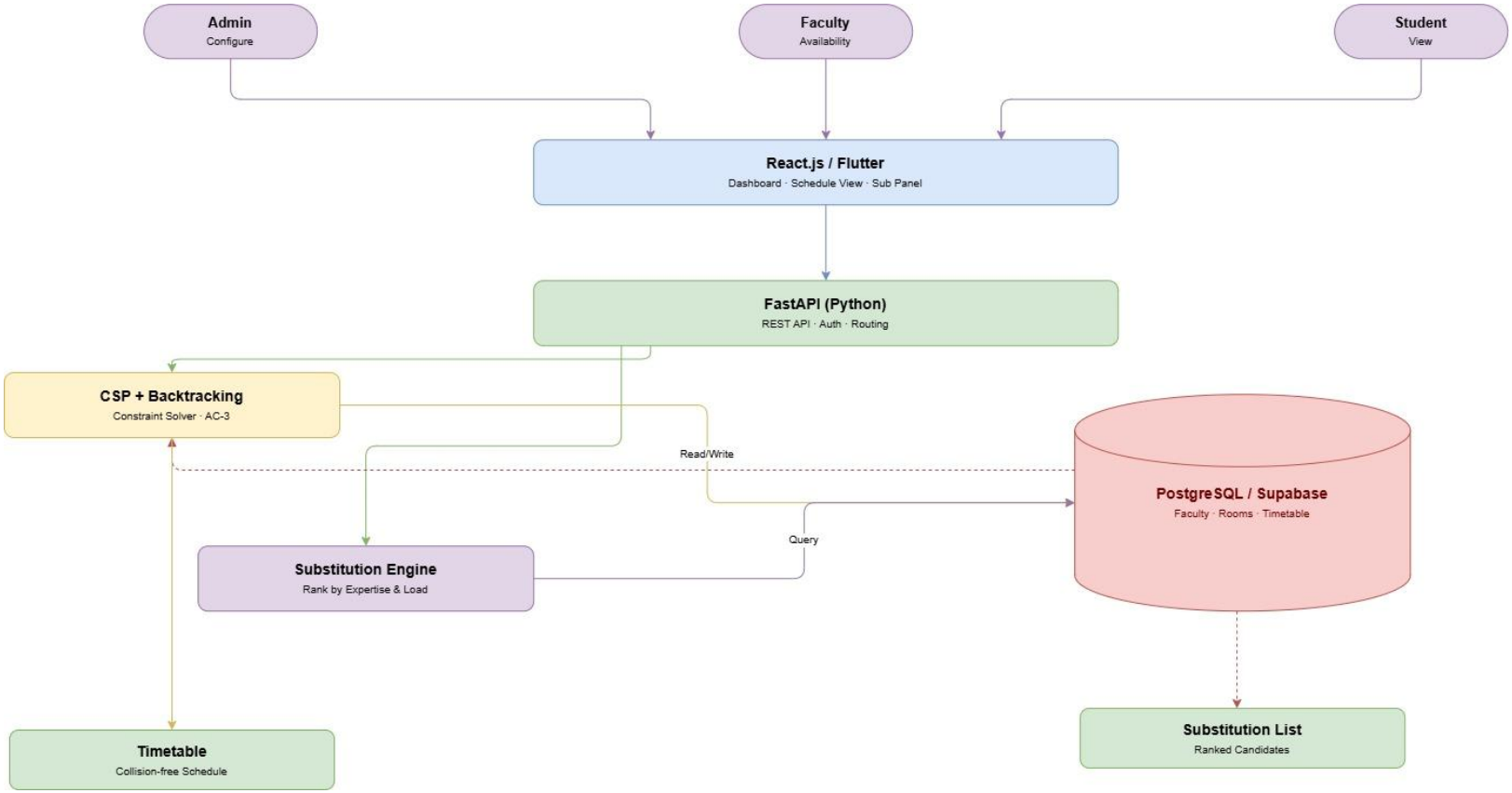
Proposed System

The proposed STTG system provides rapid, automatic schedule generation powered by a CSP algorithm with backtracking, which systematically explores all scheduling parameters to guarantee a collision-free timetable. It instantly flags and prevents any time or resource overlaps before they occur through built-in conflict detection, while a Smart Substitution module automatically identifies substitute teachers by querying the database for availability, subject expertise, and weekly workload balance. The system ensures an equitable distribution of lectures across faculty, supports multi-tenant administration where each admin manages their own isolated data, and maintains a full timetable history with draft and published states for seamless academic scheduling management.

Workflow


User Login
Admin/Faculty/Student logs into the system
Select Role & Module
User selects their role and desired action
Is Schedule Generated?
Yes
No
Enter Scheduling Inputs
Admin inputs Faculty, Rooms, Subjects & Timeslots
Generate Timetable
System runs CSP algorithm and produces schedule
View Timetable
User views the collision free generated timetable
Faculty Absent
No
Yes
Request Substitutuion
Admin triggers smart substitution module
System Assigns Substitute
Best-match faculty auto-assigned based on workload
Confirm and Notify
Changes saved and all users are notified

System Architecture



System Requirements

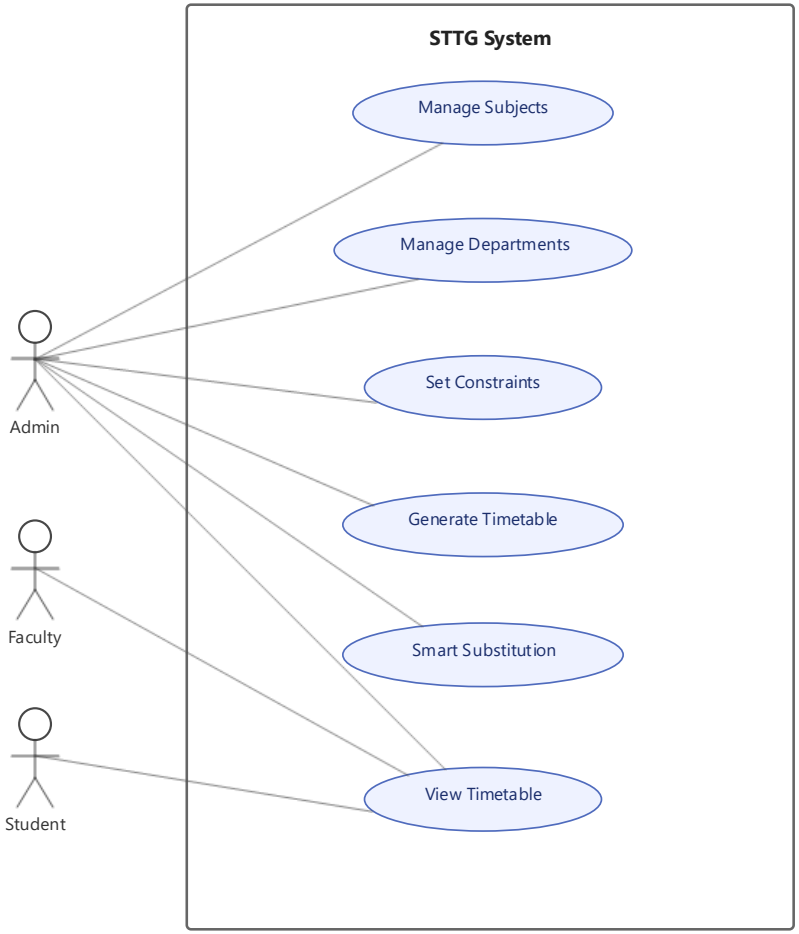
Software Requirements:

- Frontend Environment: React.js / Node Package Manager (NPM)
- Backend Environment: Python 3.x
- Database: PostgreSQL
- IDE: VS Code or PyCharm

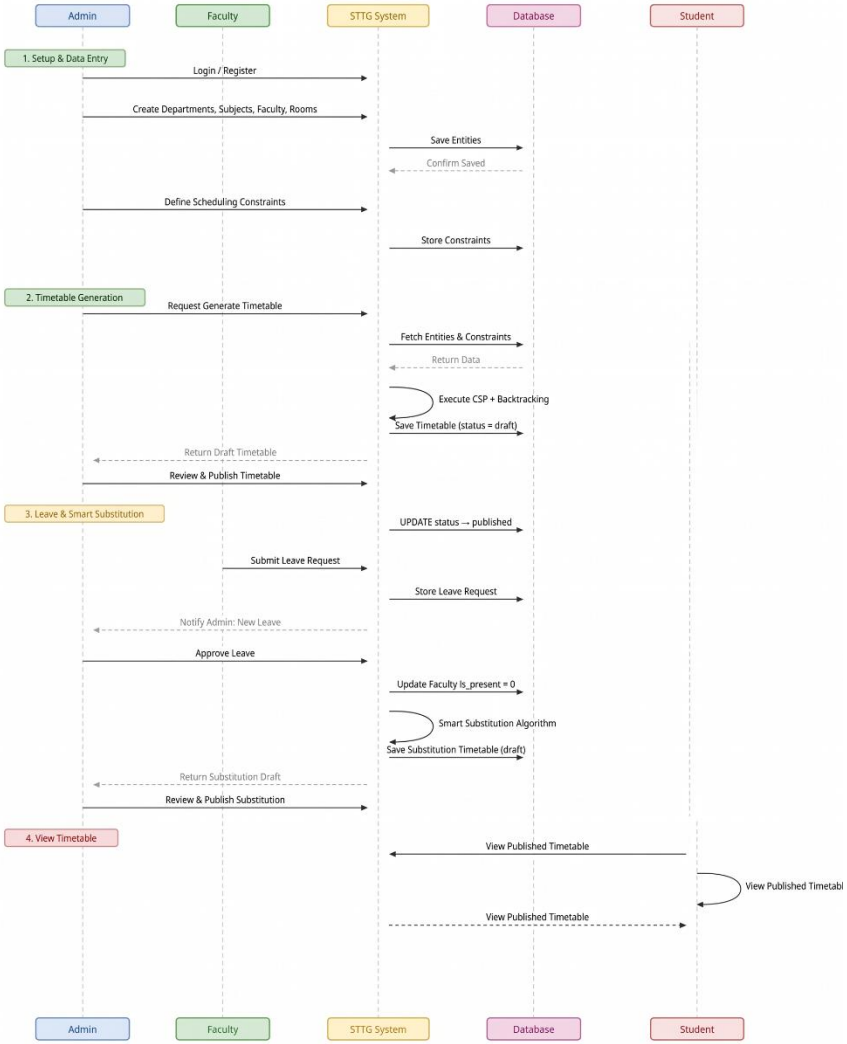
Hardware Requirements (Minimum):

- Processor: Intel Core i3 / AMD Ryzen 3 or equivalent
- RAM: 8 GB
- Storage: 256 GB SSD
- Internet: Stable connection for cloud integration

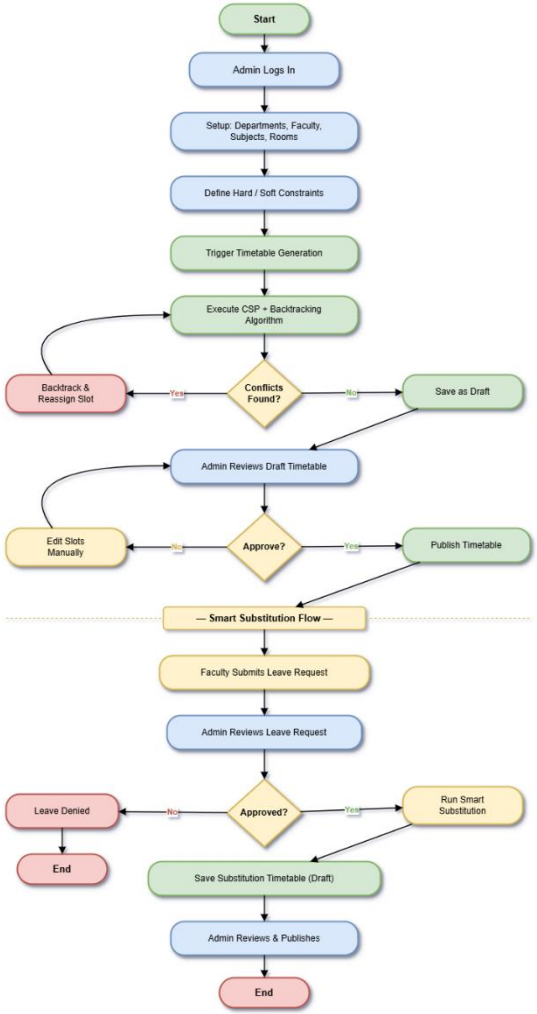
UML Diagrams –Use Case



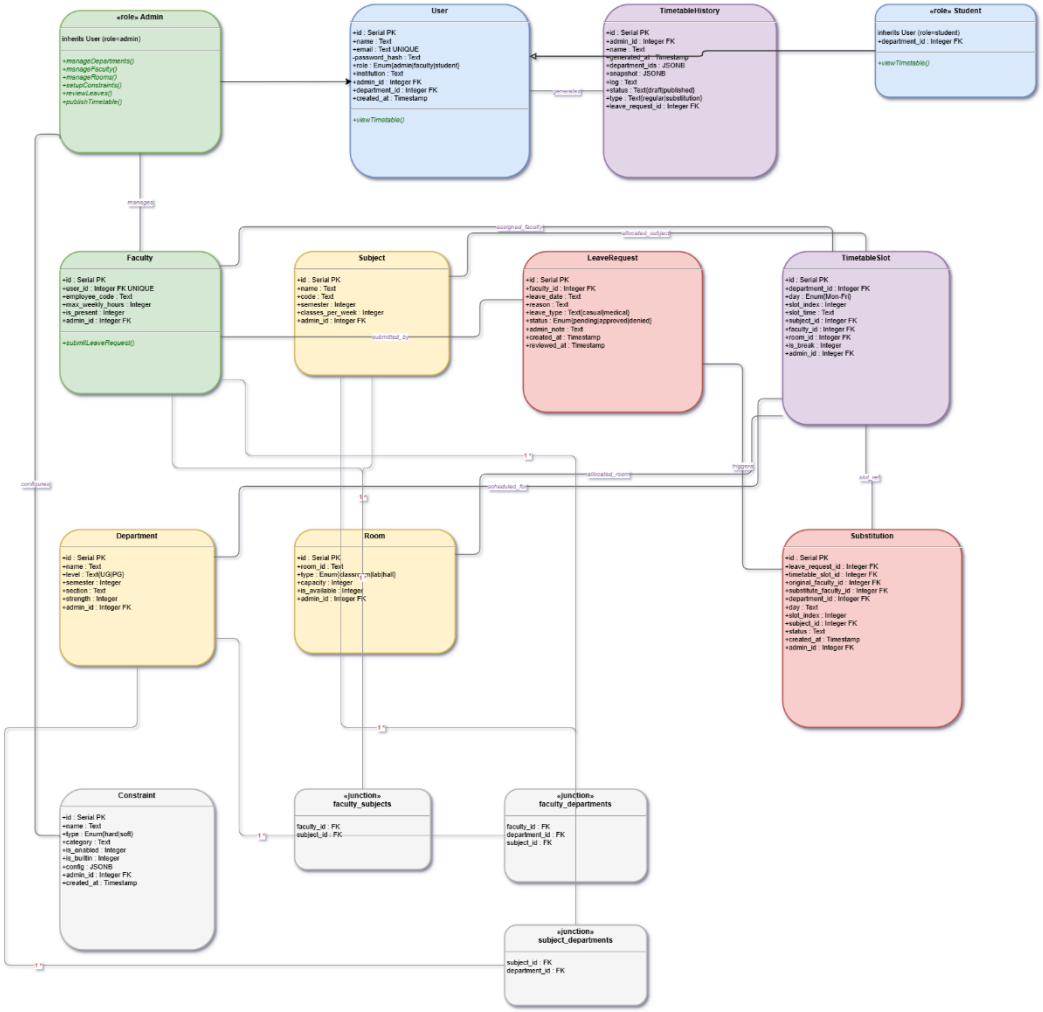
UML Diagram - Sequence



UML Diagram - Activity



UML Diagram – Class Diagram



Modules

- Data Management: Subjects, Faculty, Departments.
- Constraints Engine: Hard/Soft rule enforcement.
- Generation: CSP-based optimization.
- Smart Substitution: Backup faculty mapping.

References

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Conclusion

In conclusion, the Smart Time Table Generator (STTG) successfully addresses the complex, time-consuming challenge of academic scheduling by transforming a manual, error-prone process into an automated, highly efficient system. By leveraging advanced optimization algorithms alongside a robust, modern technology stack, STTG delivers a scalable, multi-tenant platform that guarantees clash-free timetables, seamlessly handles dynamic constraints, and intelligently manages faculty substitutions. Ultimately, STTG empowers educational institutions to minimize administrative overhead, optimize resource allocation, and ensure a smooth, uninterrupted academic experience for both faculty and students.